

High fidelity digital inline holographic method for 3D flow measurements

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Abstract: Among all the 3D optical flow diagnostic techniques, digital inline holographic particle tracking velocimetry (DIH-PTV) provides the highest spatial resolution with low cost, simple and compact optical setups. Despite these advantages, DIH-PTV suffers from major limitations including poor longitudinal resolution, human intervention (i.e. requirement for manually determined tuning parameters during tracer field reconstruction and extraction), limited tracer concentration, and expensive computations. These limitations prevent this technique from being widely used for high resolution 3D flow measurements. In this study, we present a novel holographic particle extraction method with the goal of overcoming all the major limitations of DIH-PTV. The proposed method consists of multiple steps involving 3D deconvolution, automatic signal-to-noise ratio enhancement and thresholding, and inverse iterative particle extraction. The entire method is implemented using GPU-based algorithm to increase the computational speed significantly. Validated with synthetic particle holograms, the proposed method can achieve particle extraction rate above 95% with fake particles less than 3% and maximum position error below 1.6 particle diameter for holograms with particle concentration above 3000 particles/mm³. The applicability of the proposed method for DIH-PTV has been further validated using the experiment of laminar flow in a microchannel and the synthetic tracer flow fields generated using a DNS turbulent channel flow database. Such improvements will substantially enhance the implementation of DIH-PTV for 3D flow measurements and enable the potential commercialization of this technique.

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References and links

1. T. Hori and J. Sakakibara, "High-speed scanning stereoscopic PIV for 3D vorticity measurement in liquids," *Meas. Sci. Technol.* **15**(6), 1067–1078 (2004).
2. C. E. Willert and M. Gharib, "Three-dimensional particle imaging with a single camera," *Exp. Fluids* **12**(6), 353–358 (1992).
3. G. E. Elsinga, F. Scarano, B. Wieneke, and B. W. Van Oudheusden, "Tomographic particle image velocimetry," *Exp. Fluids* **41**(6), 933–947 (2006).
4. J. Katz and J. Sheng, "Applications of holography in fluid mechanics and particle dynamics," *Annu. Rev. Fluid Mech.* **42**(1), 531–555 (2010).
5. J. Sheng, E. Malkiel, and J. Katz, "Using digital holographic microscopy for simultaneous measurements of 3D near wall velocity and wall shear stress in a turbulent boundary layer," *Exp. Fluids* **45**(6), 1023–1035 (2008).
6. S. Talapatra and J. Katz, "Three-dimensional velocity measurements in a roughness sublayer using microscopic digital in-line holography and optical index matching," *Meas. Sci. Technol.* **24**(2), 024004 (2013).
7. L. Tian, N. Loomis, J. A. Domínguez-Caballero, and G. Barbastathis, "Quantitative measurement of size and three-dimensional position of fast-moving bubbles in air-water mixture flows using digital holography," *Appl. Opt.* **49**(9), 1549–1554 (2010).
8. D.R. Guildenbecher, J. Gao, P.L. Reu, and J. Chen, "Digital holography reconstruction algorithms to estimate the morphology and depth of nonspherical absorbing particles," in *SPIE Optical Engineering and Applications* (ISOP, 2012), paper 849303.

9. F. Soulez, L. Denis, C. Fournier, E. Thiébaud, and C. Goepfert, "Inverse-problem approach for particle digital holography: accurate location based on local optimization," *J. Opt. Soc. Am. A* **24**(4), 1164–1171 (2007).
10. H. Meng, G. Pan, Y. Pu, and S. H. Woodward, "Holographic particle image velocimetry: from film to digital recording," *Meas. Sci. Technol.* **15**(4), 673–685 (2004).
11. J. Goodman, *Introduction to Fourier Optics* (McGraw-Hill, 1968).
12. D. A. Agard, "Optical sectioning microscopy: cellular architecture in three dimensions," *Annu. Rev. Biophys. Bioeng.* **13**(1), 191–219 (1984).
13. T. Latychevskaia, F. Gehri, and H.-W. Fink, "Depth-resolved holographic reconstructions by three-dimensional deconvolution," *Opt. Express* **18**(21), 22527–22544 (2010).
14. L. Dixon, F. C. Cheong, and D. G. Grier, "Holographic deconvolution microscopy for high-resolution particle tracking," *Opt. Express* **19**(17), 16410–16417 (2011).
15. T. Latychevskaia and H. W. Fink, "Holographic time-resolved particle tracking by means of three-dimensional volumetric deconvolution," *Opt. Express* **22**(17), 20994–21003 (2014).
16. D. K. Singh and P. K. Panigrahi, "Three-dimensional investigation of liquid slug Taylor flow inside a micro-capillary using holographic velocimetry," *Exp. Fluids* **56**(1), 6–15 (2015).
17. J. Sheng, E. Malkiel, and J. Katz, "Digital holographic microscope for measuring three-dimensional particle distributions and motions," *Appl. Opt.* **45**(16), 3893–3901 (2006).
18. G. E. Elsinga, "Complete removal of ghost particles in Tomographic-PIV," in 10th International Symposium on Particle Image Velocimetry (KSOV, 2013), pp. 1–3.
19. H. Meng, W. L. Anderson, F. Hussain, and D. D. Liu, "Intrinsic speckle noise in in-line particle holography," *J. Opt. Soc. Am. A* **10**(9), 2046–2058 (1993).
20. M. Malek, D. Allano, S. Coëtmelec, and D. Lebrun, "Digital in-line holography: influence of the shadow density on particle field extraction," *Opt. Express* **12**(10), 2270–2279 (2004).
21. Y. Zhang, G. Shen, A. Schrder, and J. Kompenhans, "Influence of some recording parameters on digital holographic particle image velocimetry," *Opt. Eng.* **45**(7), 075801 (2006).
22. F. Soulez, L. Denis, C. Fournier, E. Thiébaud, and C. Goepfert, "Inverse-problem approach for particle digital holography: accurate location based on local optimization," *J. Opt. Soc. Am. A* **24**(4), 1164–1171 (2007).
23. J. Gao, "Development and applications of digital holography to particle field measurement and in vivo biological imaging," PhD diss. Purdue University (2014).
24. H. Royer, "An application of high-speed microholography: the metrology of fogs," *Nouv. Rev. Opt.* **5**(2), 87–93 (1974).
25. J. C. Crocker and D. G. Grier, "Methods of digital video microscopy for colloidal studies," *J. Colloid Interface Sci.* **179**(1), 298–310 (1996).
26. Y. Li, E. Perlman, M. Wan, Y. Yang, R. Burns, C. Meneveau, R. Burns, S. Chen, A. Szalay, and G. Eyink, "A public turbulence database cluster and applications to study Lagrangian evolution of velocity increments in turbulence," *J. Turbul.* **9**(31), 1–30 (2008).
27. J. Graham, K. Kanov, X. I. A. Yang, M. K. Lee, N. Malaya, C. C. Lalescu, R. Burns, G. Eyink, A. Szalay, R. D. Moser, and C. Meneveau, "A Web Services-accessible database of turbulent channel flow and its use for testing a new integral wall model for LES," *J. Turbul.* in press.

1. Introduction

Particle Image Velocimetry (PIV) is arguably the most popular technique for whole-field velocity measurements. However, its implementation in three dimensions (3D) faces many challenges and remains an active research area. The widely-used 3D PIV techniques, including scanning PIV [1], defocusing PIV [2], and tomographic PIV [3] are still insufficient in their ability to resolve detailed turbulent structures and flows near solid boundaries due to the limited spatial or temporal resolutions. Recently, particle tracking velocimetry based on digital inline holography (DIH), namely DIH-PTV, allows direct integration with microscopic setup, which provides superior resolution for 3D flow measurements with the relevant scales ranging from μm to sub-mm [4–6]. A standard plane-wave DIH-PTV setup consists of a single camera with an imaging lens, a laser source, and other collimating optics (Fig. 1). The camera records the light interference from the forward scattering of tracer particles and unscattered portion of the laser into holograms. The holograms are then digitally reconstructed into 3D optical fields, from which the 3D tracer positions are extracted. Tracking algorithms are subsequently applied to determine tracer displacements and 3D velocity fields. Compared with other 3D PIV techniques, the DIH-PTV provides a low-cost solution for 3D flow measurements, which requires significantly less laser power for illumination and yields simple calibration and data acquisition.

For DIH-PTV implementation, the most critical and the most challenging step of DIH-PTV is to extract 3D location of tracer particles from the recorded holograms. For DIH in general, the object extraction methods can be divided into three categories as follows:

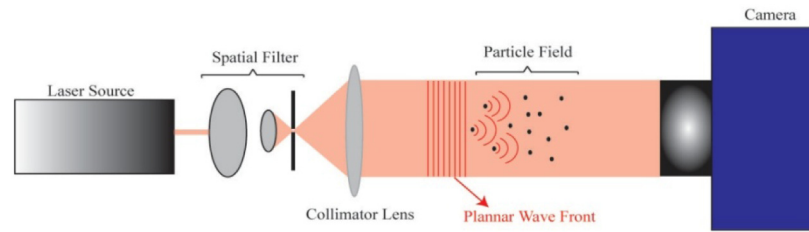


Fig. 1. The optical setup of digital in-line holography.

(a) *Focus metric based on 2D projection* (e.g [7, 8].): The 2D minimum intensity map of reconstructed 3D image is used to detect objects and determine their lateral positions, and in-depth location of the objects are determined by different focus metrics such as the intensity distribution, the sharpness, etc. This approach uses 2D image for object segmentation, which significantly improves the computational speed and lowers the memory requirements of DIH-PTV. However, using 2D projection for particle detection is susceptible to the noise generated by the cross-interference of laterally-adjacent particles, which can produce a large number of fake particles as the particle concentration increases. In addition, the longitudinal resolution of this approach relies on accurate detection of particle edges. Large uncertainty can be generated when applying this approach to smaller objects such as flow tracers of typically around 2-3 pixel in diameter. For the techniques that employ minimum intensity as a focus metric (e.g [7].), a user defined intensity threshold is used to detect and segment objects, which is subjected to changes under different hologram recording conditions.

(b) *Inverse simulation and residual calculation* (e.g [9].). The inverse approach first generates a modeled hologram based on diffraction theorem using prescribed parameters (e.g. 3D location and object size, etc.). Then, through an iterative process, these parameters are being gradually refined to minimize the difference between the modeled hologram and the recorded one. The refined parameter at the end provides an estimate of object location and size. Although the inverse approach can provide subpixel resolution in the longitudinal direction, it is susceptible to the noises from the background and the cross-interference of objects, which is not suitable for applications with dense tracer holograms. In addition, the iterative modeling of holograms by scanning through several in-depth positions and calculation of residual hologram leads to a highly expensive set of computations.

(c) *3D segmentation* (e.g [5, 6, 10].): It uses reconstructed 3D optical fields to simultaneously detect objects and calculate their 3D locations. In this approach, each cross section of reconstructed 3D optical fields is first segmented based on user defined intensity threshold. Then, the 2D segments of each object in adjacent cross sections are joined together to form 3D elongated objects, and the 3D centroid location of each object is determined using geometric average or intensity-weighted average of the location of each voxel within the object. Compared with focus metric based on 2D projection, this approach can work with higher concentration for small tracer particles, and it reduces the uncertainty on particle detection significantly. However, this approach still suffers from poor longitudinal resolution associated with depth-of-focus (DOF) [4] issue, and requires substantial human intervention to optimize the intensity threshold for segmentation. It is also computationally expensive compared with the focus metric approach.

In summary, both focus metric and inverse simulation methods have stringent requirements on object geometry, limiting their use in the concentrated fields of particle tracers of 2-3 pixel in diameter. Although 3D segmentation approach can work with holograms with relatively higher concentration of objects, the effectiveness of this method

still deteriorates rapidly with increasing particle concentration due to the noise from out-of-plane particles. In addition, the current 3D segmentation algorithm faces other major challenges including poor longitudinal resolution, substantial human intervention, and expensive computation. Despite the advantages of DIH-PTV, these challenges significantly limit its implementation for 3D flow measurements, unlike some other commercialized 3D PIV techniques, e.g. defocusing PIV and Tomographic PIV.

In the current study, we introduce a novel holographic particle extraction method that overcomes the above mentioned challenges and makes DIH-PTV a mature technique for 3D flow measurements. The paper is structured as follows. Sec. 2 provides a detailed description of each step of our particle extraction method. The effectiveness and efficiency of this method is evaluated using synthetic and experimental flows with tracer particles in Sec. 3. The paper concludes with a summary and discussion in Sec. 4.

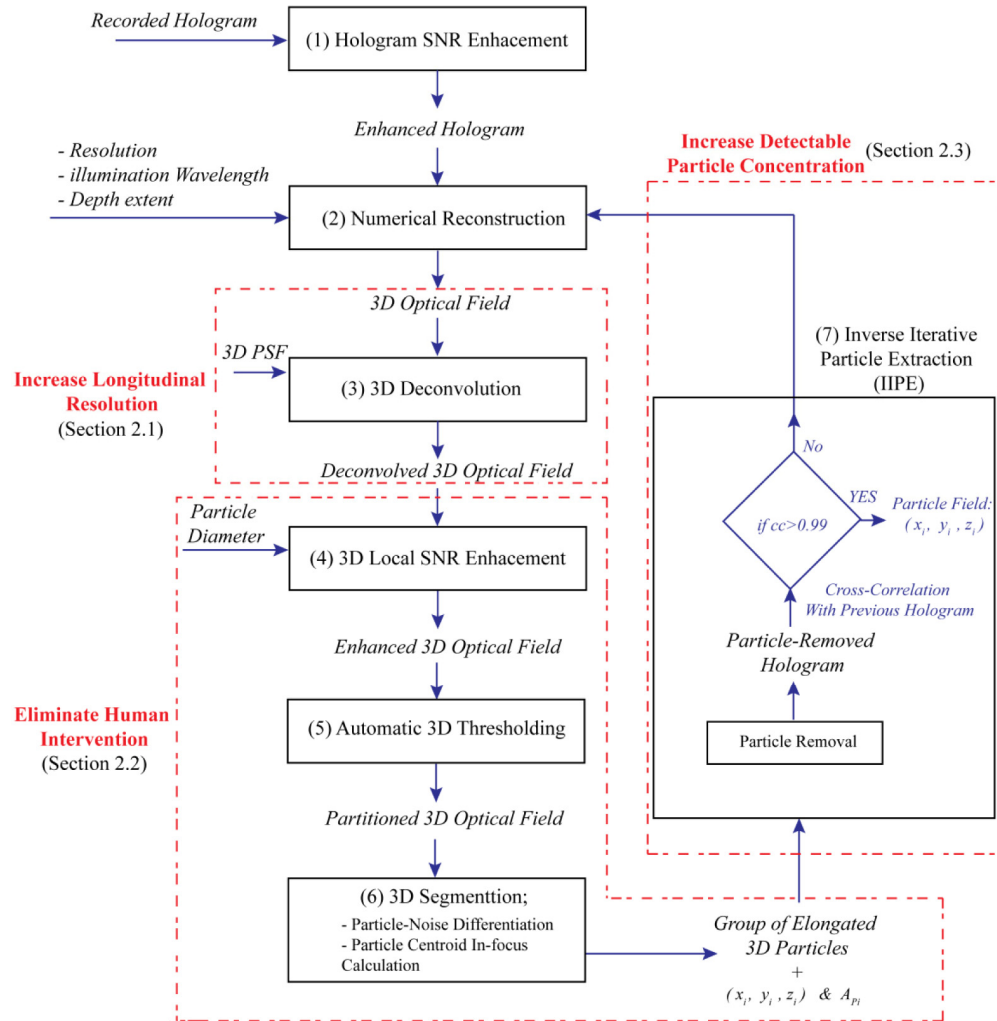


Fig. 2. The flow chart of our DIH-PTV particle extraction procedure.

2. Methodology

As illustrated in Fig. 2, the proposed method for particle extraction consists of seven major steps including: (1) hologram enhancement; (2) numerical reconstruction; (3) 3D

deconvolution; (4) 3D local SNR enhancement; (5) automatic thresholding; (6) 3D segmentation; and (7) inverse iterative particle extraction (IIPE). The hologram enhancement is conducted through subtraction of holograms from corresponding time-averaged hologram followed by gray-scale equalization. In addition, to increase the longitudinal resolution of DIH, we enriched our holographic processing algorithms with a 3D deconvolution technique (Step No.3). To minimize human intervention and enable automatic particle extraction, we develop an automatic 3D local SNR enhancement and thresholding algorithm for 3D segmentation (Step No.4-6). IIPE (Step No.7) is introduced to maximize high fidelity particle extraction for concentrated particle tracer holograms. Finally, the entire DIH processing is accelerated using graphics processing units (GPUs), which reduces the computation time required for DIH processing (e.g. ~2–5 minutes, depending on number of particles, for a $2k \times 2k$ holograms with 512 longitudinal scans on a system with an Intel Core i5-4690S 3.20 GHz CPU, a GeForce GTX 970 GPU and a 16 GB RAM). Below, we will present a detailed description of the abovementioned new ingredients introduced for DIH-PTV processing including their mathematical frameworks, algorithms and the test results using holographic data from experiments.

2.1 Increasing longitudinal resolution: 3D deconvolution

To increase the longitudinal resolution of DIH-PTV, we adapt the computational deconvolution approach, used in conventional fluorescence microscopy, for processing tracer particle holograms. For completeness, we will first briefly describe the reconstruction process of DIH, and then discuss the DIH deconvolution approach in detail.

In DIH imaging, by treating the recorded hologram as a 2D aperture the 3D image of the objects (the 3-D complex optical field) can be reconstructed through:

$$u_p(x, y, z) = I_h(x, y) \otimes h(x, y, z) \quad (1)$$

where u_p is the reconstructed 3D optical field, x, y , and z are lateral and longitudinal locations respectively, $\otimes$ represents the convolution operator and $h(x, y, z)$ is the point-spread function (PSF) introduced by diffraction. This diffraction PSF is modeled using the Rayleigh-Sommerfield Kernel [11]:

$$h(x, y, z) = \frac{1}{j\lambda\sqrt{x^2 + y^2 + z^2}} \exp[jk(\sqrt{x^2 + y^2 + z^2})] \quad (2)$$

where k is the propagation vector and λ is the wavelength length of illumination beam. In order to accelerate the computation, the convolution integral is usually calculated as a simple multiplication in the Fourier domain using fast Fourier transform as below

$$u_p(x, y, z) = FFT^{-1}\{FFT[I_h(x, y)] \times FFT[h(x, y, z)]\} \quad (3)$$

where $FFT[]$ represents the fast Fourier transform operator. In brief, through convolving the recorded hologram ($I_h(x, y)$) with the diffraction PSF ($h(x, y, z)$), the corresponding 3D optical field including stack of longitudinal scans is reconstructed. This optical field represents the 3D image of the corresponding particle field, which suffers from DOF problem and noises associated with the cross-interference of adjacent objects and out-of-focus objects. Similar problem is present in another linear and space-invariant imaging system, i.e. fluorescence microscopy, where the volumetric information of objects is captured on a stack of imaging planes at different longitudinal positions. In fluorescence microscopy, this problem was first tackled through a computational image restoration technique called deconvolution by Agard [12]. The 3D deconvolution based on Wiener filtering formulation was successfully adapted for DIH images through instant 3D deconvolution, and was proved to reduce the DOF and consequently increase the longitudinal resolution significantly [13–15]. An iterative scheme

for 3D deconvolution of holograms was also proposed by Litychevskaya *et al.* [13], which requires larger memory and higher computational cost. We have tested the performance of both instant and iterative deconvolution schemes for concentrated particle fields, which shows that the instant scheme detects higher number of particle traces with lower SNRs compared to the iterative scheme. Therefore, to extract more particles with a feasible computational time, we employ instant 3D deconvolution for DIH-PTV processing. The drawback of a lower SNR compared to that of the iterative approach is also resolved in the subsequent processing steps of our method.

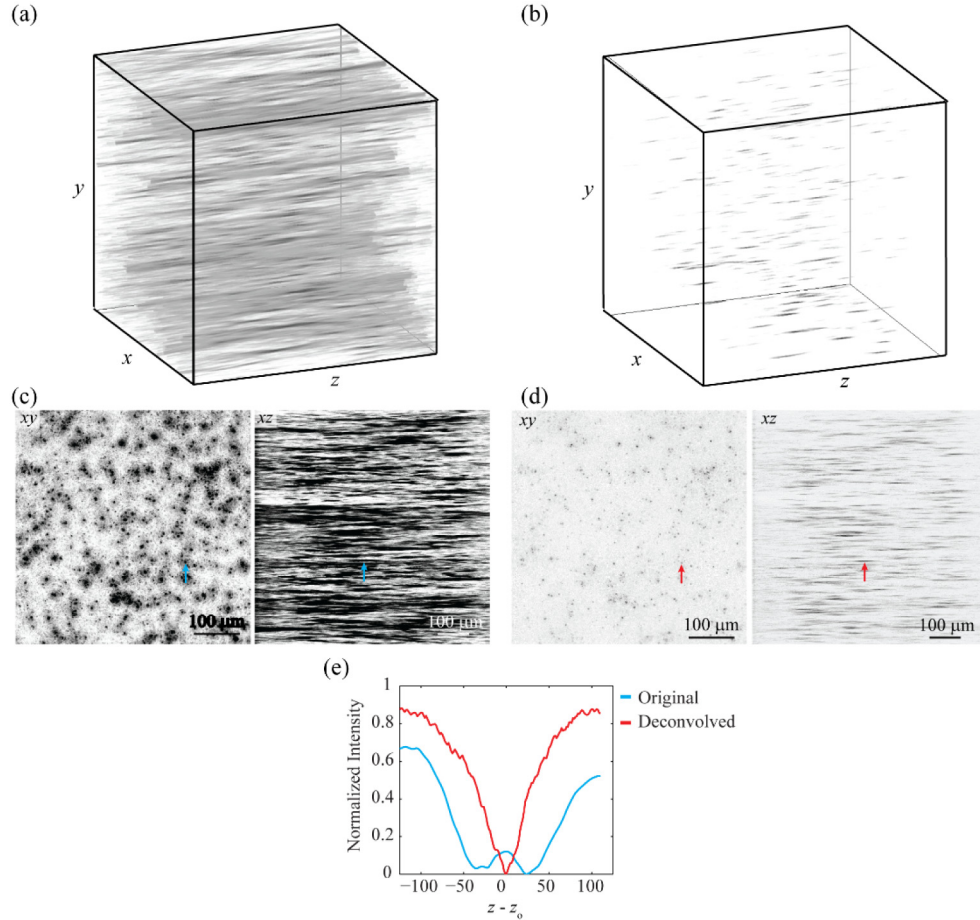


Fig. 3. (a) Original and (b) deconvolved 3D reconstruction of a sample hologram of 2 μm -diameter silver coated silica tracer particles. (c) The xy and xz plane minimum intensity maps of (c) the original and (d) the deconvolved 3D optical fields. (e) A normalized longitudinal intensity profile of a sample particle before and after deconvolution at the location pointed by the arrow. Note that the intensity values at different longitudinal locations presented here are the mean values averaged over the lateral cross-sections of the particle.

The instant deconvolution scheme computes the deconvolved 3D optical field as follows:

$$u'_p(x, y, z) = FFT^{-1} \left\{ \frac{FFT[I_{PSF}(x, y, z)]^* \times FFT[I_p(x, y, z)]}{FFT[I_{PSF}(x, y, z)] \times FFT[I_{PSF}(x, y, z)]^* + \beta} \right\} \quad (4)$$

where I_{PSF} and I_p represent the intensity distributions of PSF of the optical system and the original reconstructed 3D optical field, respectively, u'_p is the corresponding deconvolved

field, and β is a small constant used to prevent probable divisions by zero. The value of β should be smaller than the magnitude of the other term in the denominator, and is selected as ~ 0.5 for a wide range of tested tracer holograms. The 3D PSF function of the optical system (I_{PSF}) is modeled, using Rayleigh-Sommerfield's diffraction, as a 3D reconstruction of a synthetic hologram generated for a one pixel size aperture located in the center of measurement volume. It is noteworthy that the 3D PSF function can be also obtained experimentally through reconstruction of a hologram of a point-like object.

To demonstrate the improvement using this method, the reconstructed 3D optical fields from a tracer field hologram before and after deconvolution are presented in Fig. 3. The hologram is obtained from a flow experiment in a $1 \times 1 \text{ mm}^2$ cross-section microchannel operated with water seeded with $2 \text{ }\mu\text{m}$ silver coated silica particles. As Fig. 3 shows, the DOF of the particle field is reduced $\sim 20\%$ – 40% of its original length after deconvolution. In addition, the distinction between in-focus particles and the background noises from out-of-focus particles is improved substantially, as illustrated in Fig. 3(c) and Fig. 3(d). However, despite the significant improvement of the longitudinal and lateral resolution, large variation in the intensity distribution of particles and the SNR still remains across the reconstructed 3D optical field after deconvolution, similar to that in the original reconstruction. This variation of SNR is caused by several factors including inhomogeneous tracer concentration in the measurement volume, e.g. variation of particle-particle lateral closeness, and variations in particle size in the measurement volume, etc. In addition, the current deconvolution process inevitably reduces or even eliminates the signature of tracer particles close to the borders of measurement volume or whose fringes are contaminated by neighboring objects. This drawback results in significantly less detectable tracers, and subsequently reduces the spatial resolution of vector fields for DIH-PTV. These issues that are remained after deconvolution or induced by deconvolution will be addressed in subsequent sections 2.2 and 2.3.

2.2 Eliminating human intervention: automatic 3D SNR enhancement, thresholding and segmentation

To minimize human intervention and consequently reduce measurement uncertainty during particle extraction, we develop a high-fidelity automatic process to segment 3D particles from a reconstructed 3D optical field. As mentioned in Sec. 2.1, a large variation of SNR still remains across the reconstructed 3D optical field after deconvolution. Therefore, a single value of intensity threshold (e.g [8].) is not adequate for segmenting tracer particles effectively in DIH-PTV applications, leading to different degrees of oversampling or under-sampling of objects. To overcome this issue, Singh and Panigrahi [16] proposed a 2D automatic thresholding technique that calculates a set of different threshold values for each reconstructed cross sections. Although this approach takes into account the longitudinal variation of SNR, it is ineffective to deal with the variation of threshold due to the particle concentration inhomogeneity along the lateral directions. On the other hand, the existing approach using 3D thresholding [17] involves a number of user defined parameters and may introduce measurement uncertainty caused by human intervention. To overcome these issues, our proposed method consists of three steps. The intensity of the deconvolved 3D optical field is first equalized using a 3D local SNR enhancement. Then, a threshold calculation algorithm is applied to automatically determine the optimal intensity threshold for the entire optical field. Finally, the 3D objects are formed by joining neighboring voxels together within the thresholded optical field. The detail description of each step is provided below.

3D Local SNR Enhancement: The 3D local SNR enhancement first divides the deconvolved optical field into object domain and background using a single threshold, and then performs local intensity normalization on the object domain. The threshold for isolating background from regions with particles, I_{Thr0} , is automatically determined using 2D minimum intensity map of the optical field as follows:

$$I_{Thr0} = Avg(I_{min}) - Avg(I_{\sigma}) - \sigma(I_{\sigma}) \quad (5)$$

where $Avg()$ and $\sigma()$ are average and standard deviation operators, I_{min} and I_{σ} represent the minimum and standard deviation of intensity of pixels within an interrogation window which scans over the entire 2D minimum intensity map of the optical field with 50% overlap. The size of interrogation window is selected to be $4 \times D_p$ in order to preserve sufficient information of the particle fringes, where D_p is the particle diameter, e.g. 8×8 pixels for 2×2 pixels particles. Considering the objects (i.e. tracer particles) are dark spots in the optical field, the background is defined as the domain with voxel intensities above I_{Thr0} and these voxels are assigned with the value of the maximum intensity of the 3D optical field. The subsequent local intensity normalization is conducted through 3D min-max filtering over the object domain as below:

$$I'(x, y, z) = \frac{I(x, y, z) - Min(I_v)}{Max(I_v) - Min(I_v)} \quad (6)$$

where $I(x, y, z)$ and $I'(x, y, z)$ are the original and the normalized intensity values of a voxel located at (x, y, z) , respectively, and the $Min()$ and $Max()$ are the minimum and maximum calculation operators, respectively, and I_v represents the intensity distribution within the corresponding interrogation block shifting over the entire object domain with 50% overlap. The size of these blocks is determined using the DOF estimate of particle objects through synthetic hologram simulation with similar magnification and for largest tracer used in the experiment. The DOF is defined as the distance between the 75% intensity peaks around the particle centroids [17], and to account for the noises present in the actual experiment, we double the synthetically calculated DOF as an estimate of DOF. This estimate is in a good agreement with the available experimental data of similar magnifications and particle sizes (e.g [17].).

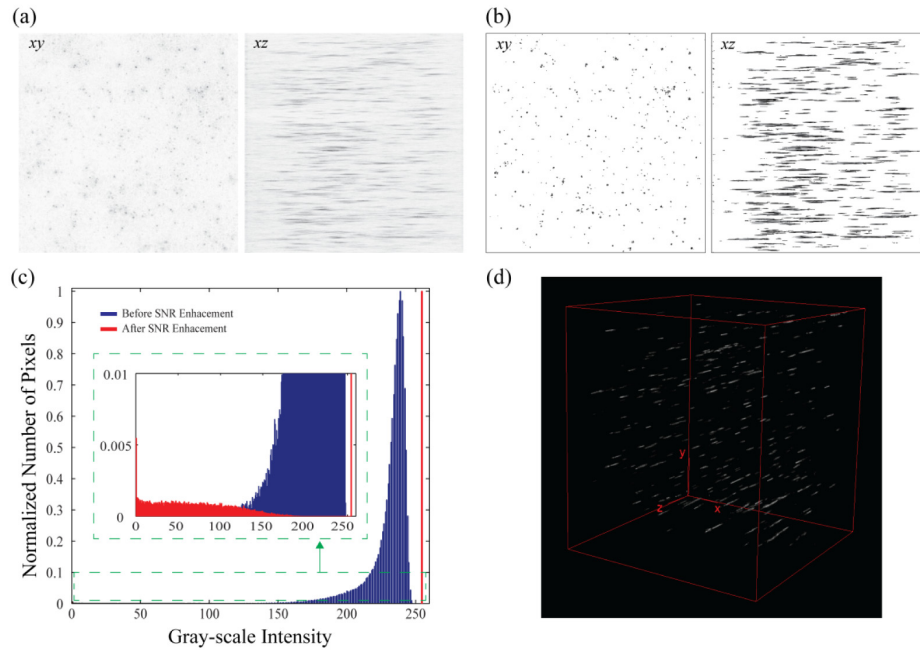


Fig. 4. The xy and xz plane minimum intensity maps of the deconvolved 3D reconstruction (a) before and (b) after 3D local SNR enhancement, for $C_p \approx 3000$ particles/mm³. (c) The intensity distribution profiles of minimum intensity images after and before 3D local SNR enhancement for (d) corresponding thresholded and 3D segmented particle field.

To demonstrate the effectiveness of this method, Figs. 4(a) and 4(b) show the xy and xz minimum intensity maps from the same 3D optical field used in Fig. 3, before and after 3D local SNR enhancement. As the figures show, the uniformity of particle intensity is improved and the background noise is suppressed significantly after 3D SNR enhancement, which facilitates the subsequent processing steps. It is noteworthy that particle concentrations, C_p , for experimental holograms are calculated from the tracking trajectories of particles in time, i.e. only those particles are counted real that are present in trajectories with matching pairs from five consecutive holograms.

Automatic thresholding: After 3D local SNR enhancement, a single threshold value can be applied to the entire measurement volume to segment the particles from the background. The value is automatically determined using intensity histogram of xy 2D minimum intensity map of the reconstructed 3D optical field. As shown in Fig. 4(c), a clear separation between background and objects is achieved after SNR enhancement. Therefore, the detection threshold value is automatically calculated from the histogram of intensity values of the SNR enhanced minimum intensity map by finding the maximum intensity within the object domain.

3D object segmentation: After automatic thresholding, the 3D object segmentation algorithm joins the 2D segments into 3D objects through merge operator connecting thresholded voxels located within certain distance from each other as shown in Fig. 4(d). Through preliminary testing, the distance is selected to be $2 \times D_p$. To reduce the false positives during particle segmentation, a subsequent size filter is applied to eliminate false particles. It is noteworthy that unlike tomographic PIV where the size of the reconstructed particles does not reveal an appreciable difference between fake and real particles [18], there is a distinction between particles and noises in DIH-PTV. The fake particles resulted from cross-interference or out-of-focus diffraction do not exhibit clear longitudinal elongation, and therefore yield considerably smaller longitudinal length compared to real ones. The lowest estimate of DOF for the particle field is employed in order to perform this filtration on detected 3D blobs, i.e. particles are those blobs with longitudinal length longer or equal to this DOF length. This length criterion is automatically calculated through a synthetic hologram simulation with similar magnification and for smallest tracer size used in the experiment. Particle centroids (x_i, y_i, z_i), diameters (D_{pi}) and in-focus cross sections (A_{pi}) are finally calculated based on geometrical center for filtered 3D blobs. A velocity vector field is calculated for any two consecutive holograms using their corresponding extracted centroid fields (i.e. x_i, y_i, z_i).

2.3 Increasing detectable particle concentration: iterative inverse particle extraction

As mentioned in Sec. 2.1, the current 3D deconvolution process inevitably reduces or eliminates the signature of particle objects, specifically for those contaminated by cross-interference and those located close to the borders of reconstructed 3D optical field. The latter case results in truncation of the signal from 3D particle image, which reduces its correlation with the prescribed PSF [15]. These issues significantly limit the concentration of particles that can be segmented effectively.

To increase the concentration of detectable particles, an algorithm is developed to keep updating the reconstructed 3D optical field by removing extracted particles from it and conduct the aforementioned particle extraction steps on the updated optical field in an iterative fashion. This algorithm, namely iterative inverse particle extraction (IIPE), can accentuate the fringe patterns of the previously contaminated/hidden particles, allowing higher concentration of particles to be extracted. Specifically, each extracted particle is removed from the reconstructed optical field by filling its in-focus cross section with the average complex value of the reconstructed optical field (background). The updated optical field (with extracted particles signature removed) is then used to generate the updated

hologram through backward diffraction for the next round of particle extraction. This iterative process can be formulated in following steps:

$$u_p(x, y, z_i)_n = I'_h|_n(x, y) \otimes h(x, y, z_i) \quad (7)$$

$$u_p(A_{pi}(x_i, y_i), z_i)_n = \text{Avg}(u_p(x, y, z_i)_n) \quad (8)$$

$$I'_h|_n(x, y) = |u_p(x, y, z_i)_n \otimes h(x, y, -z_i)|^2 \quad \& \quad i = i + 1 \quad (9)$$

where i represents the particle ID number ranging from 1 to N_p (the total number of detected particles at removal iteration n), A_{pi} is the in-focus cross section of particle i (calculated in 3D object segmentation step), and $I'_h|_n$ and $u_p(x, y, z_i)_n$ represent the particle removed hologram and the corresponding particle-removed optical field after n iterations, respectively. Note that the order of particles does not matter in our particle removal process, and each iteration ends when all the detected particles from previous iteration are removed, i.e. $i = N_p$. The flowchart of IIPE algorithm is also depicted in Fig. 5(a).

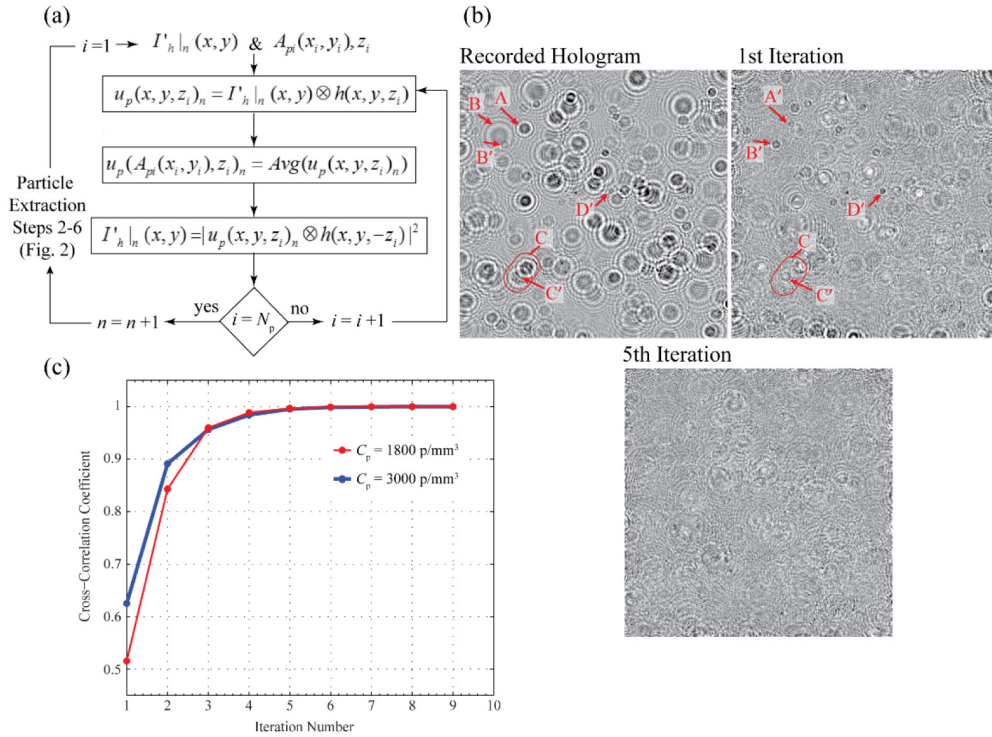


Fig. 5. (a) The IIPE algorithm flowchart. (b) Iterative particle-removal (for hologram $C_p = 1800$ particles/mm³): recorded and the corresponding particle-removed holograms after different iterations. To provide examples of the particle removal, the letter symbols without prime are used to mark a few particles with high SNR that are extracted in the first iteration. The particles marked using letter symbols with prime indicate the particles whose fringe patterns are recovered/enhanced after removing those high SNR particles in the first iteration (e.g. After removing the particle A in the first iteration, the fringes of particle A' which is laterally-close to A but at different longitudinal location appears). The corresponding results for $C_p \approx 3000$ particles/mm³ can be observed in [Visualization 1](#). (c) The cross-correlation between successive particle-removed holograms for both cases of $C_p \approx 1800$ and 3000 particles/mm³.

The iteration ends when the cross-correlation coefficient of the two consecutive updated holograms reaches 0.99, indicating that there is almost no particle that can be further extracted from the updated holograms. To illustrate this process, the iterative inverse particle extraction is implemented on the same hologram used in Fig. 3 and 4 (see Fig. 5(b) and [Visualization 1](#)). As the figure shows, substantial number of particles can be extracted through each iteration, and the entire process ends after 5 iterations when the cross-correlation coefficient exceeds the value of 0.99 (see Fig. 5(c)). Compared with single-pass particle extraction, about 30% more particles are extracted through the iterative inverse algorithm from the experimental holograms of the laminar flow within a $1 \times 1 \text{ mm}^2$ cross-section microchannel seeded with 2000-3000 particles/ mm^3 of $2 \text{ }\mu\text{m}$ particles. This result highlights the necessity and effectiveness of IIPE for DIH-PTV.

Notably, our approach of particle removal is fundamentally different from the inverse algorithm proposed by Soulez *et al.* [9]. Specifically, our approach is performed in the 3D reconstructed optical field and does not require synthesizing model holograms. In addition, the technique proposed by Soulez *et al.* [9] performs the removal on the hologram plane, which can alter the fringes of particles located laterally close to the removed particles, causing larger uncertainty and lower extraction rate for concentrated holograms. In contrast, the proposed method here enables particle removal from highly-concentrated field without deteriorating the fringes of residual particles, and thus it elevates the SNR of previously hidden/distorted particles to a sufficient level for being extracted.

3. Methodology assessment

In this section, we present the results of the implementation of our DIH-PTV method on synthetic and experimental holograms in order to evaluate its accuracy and efficiency. First, synthetic holograms are used to quantify the accuracy and robustness of our particle extraction method over a wide range of simulation parameters. Second, the fidelity of our method for 3D velocity measurements is evaluated using both the experimental data from a laminar microchannel flow and the synthetic data from direct numerical simulation (DNS) of a turbulent channel flow.

3.1 Particle extraction assessment

In inline holography, different experimental parameters such as volume depth (L_z), particle concentration (C_p) and particle size (D_p) have substantial effects on SNR, and consequently the rate and position accuracy of particle extraction [19–21]. In order to quantify these effects on the accuracy and effectiveness of our particle extraction method, we implemented the algorithm on a set of synthetic holograms generated using Rayleigh-Sommerfield diffraction theory, similar to the approach used by Zhang *et al.* [21] and Soulez *et al.* [22]. Following the approach employed by Gao [23], the 512×512 synthetic holograms are cropped out of originally simulated holograms of size 2048×2048 to reduce the aliasing effect. A total of 30 holograms are synthesized with the hologram plane to particle field distance of $600 \text{ }\mu\text{m}$, and the lateral resolution of $1 \text{ }\mu\text{m}/\text{pixel}$ and the illumination of 632 nm laser. The particle hologram includes randomly distributed monodisperse tracers with C_p ranging from 1600–4800 particles/ mm^3 . Different tracer size (i.e. $D_p = 3$ and $6 \text{ }\mu\text{m}$) and volume depth (i.e. $L_z = 0.5$ and 1 mm) are employed. After implementing our particle extraction algorithm, the extracted particle centroids are compared to their extract pairs in order to quantify the extraction rate (E_p) and position error (δ) of our algorithm for each hologram. Using the distribution of positioning errors for all particles within a hologram (positioning error for a particle is the distance between its exact and calculated centroids), δ is defined for each hologram as the maximum positioning error for individual particles which 95% of the particles lie within $-\delta$ to δ . In addition, the extraction rate is calculated for each hologram as the ratio of accurately extracted particles, i.e. the particles extracted with positioning errors of $\delta \leq 2 \times D_p$, to the total number of particles.

The results of particle extraction for all 30 holograms are presented in Fig. 6. The comparison between the particle fields extracted for holograms of $D_p = 3 \mu\text{m}$ and $L_z = 0.5 \text{ mm}$ using iterative (IIPE) and non-iterative algorithms, Fig. 6(a), shows that the E_p drops $\sim 60\%$ as the particle concentration increases from 1600 to 4800 particles/ mm^3 without IIPE, while the extraction rate stays above 97% using IIPE algorithm. It is noteworthy that the number of falsely extracted particles is less than 3% of the total number of particles using IIPE method for all of the test cases. These results highlight the effectiveness of our novel inverse iterative particle extraction method compared to the non-iterative approach, which is particularly important for DIH-PTV where the spatial resolution and accuracy of velocity measurements are directly related to extracted particle concentration. As Fig. 6(a) shows, the current algorithm can reconstruct the 3D particle fields up to $< 55 \mu\text{m}$ particle spacing within a 1 mm^3 for highly concentrated particle fields ($C_p > 3000 \text{ particles/mm}^3$).

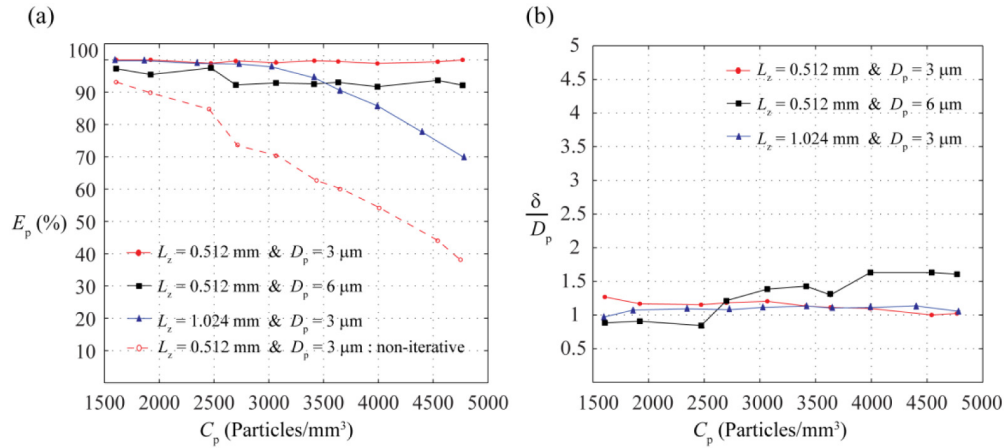


Fig. 6. The variation of (a) particle extraction rate and (b) positioning error with particle concentration, diameter and measurement volume obtained from the hologram simulation using synthetic particle fields.

It is noteworthy that the shadow density ($Sd = C_p \times D_p^2 \times L_z$ [24]), which has been commonly used to estimate the SNR and extraction efficiency (e.g [20,23].), does not serve as an appropriate parameter to evaluate the particle extraction rate of our method, which has stronger dependency on L_z than D_p . For example, comparing the case of $D_p = 6 \mu\text{m}$ and $L_z = 0.5 \text{ mm}$ with $D_p = 3 \mu\text{m}$ and $L_z = 1 \text{ mm}$, our method results in higher E_p for $D_p = 6 \mu\text{m}$ and $L_z = 0.5 \text{ mm}$ whose Sd is double of the other. Similarly, the shadow density fails to estimate the positioning error variations, since its variation depends more on D_p than L_z , as shown in Fig. 6(b). The positioning error (δ) for $D_p = 3 \mu\text{m}$ and $L_z = 0.5$ is almost the same as that of the case with $D_p = 3 \mu\text{m}$ and $L_z = 1 \text{ mm}$, while Sd for the first case is the half of the second one. This higher dependency of δ can be attributed to DOF relation with D_p (i.e. $\text{DOF} \propto D_p^2/\lambda$ [4]).

Moreover, for 3D PIV implementations, small-size particle tracers are usually employed to yield good traceability, and high tracer concentration and large sample volume are generally favored for capturing flow structures with broad dynamic range of scales. These requirements indicate the significance of both C_p and L_z , in contrast to the shadow density, for evaluating the accuracy and effectiveness of DIH-PTV.

3.2 3D velocity measurement assessment

3.2.1 Case #1: application to experimental microchannel flow

Our DIH-PTV technique is employed to quantify 3D velocity fields from the holograms recorded for a microchannel flow experiment. A motorized syringe is used to pump the low

speed ($Re = 10$, based on the channel width) water flow seeded up to $C_p \approx 3000$ particles/mm³ with 2 μ m silver coated silica particles through a 130 mm long glass microchannel with a 1 mm² square cross-section and 0.15 mm wall thickness. The digital inline holographic microscopy (DIHM) set-up contains a CMOS camera (2048 $\times$ 1088 pixels Flare 2M360-CL), a 12 mW He-Ne laser ($\lambda = 632$ nm) as the coherent light source, a spatial filter and a collimator lens. A long working distance infinity-corrected 5X objective lens (5X EO M Plan Apo) is used to record magnified holograms of the microchannel flow with 1.1 μ m/pixel lateral resolution, and the recording plane is placed at outer wall of the microchannel. A sample hologram is shown in Fig. 7(a). The 3D velocity fields are then calculated implementing an open source 3D particle tracking program [25] on the consecutive tracer fields extracted through our IPE algorithm. The process generates a series of unstructured 3D velocity fields as illustrated in Fig. 7(b). Since the flow is fully developed within the microchannel, these velocity fields are expected to replicate the parabolic profiles of a Poiseuille flow within the microchannel. The comparison between the analytical solution (Poiseuille flow profile) and the measured instantaneous velocity profile is presented in Fig. 7(d). As it shows, our DIH-PTV measurements match the analytical solution with less than a 5% difference in magnitude, which is attributed to both the velocimetry uncertainty and the discrepancy between the flow condition within the channel and those of the analytical solution. The distribution of longitudinal velocity component, Fig. 7(e), also shows that the maximum error associated with this component is less than 4% of the streamwise direction velocity magnitude.

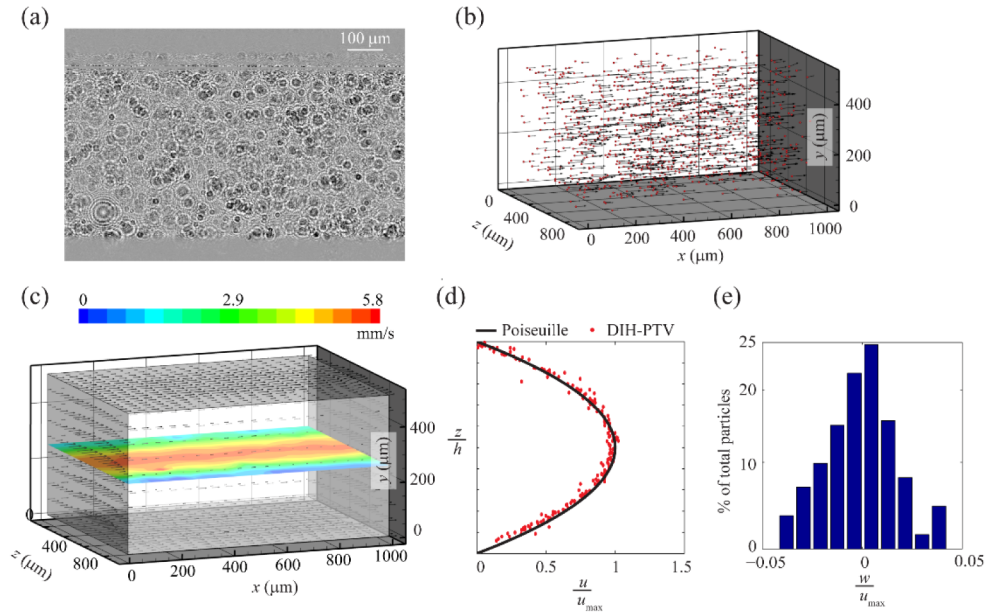


Fig. 7. (a) A sample hologram captured from a microchannel flow seeded with 2 μ m tracer particles of up to 3000 particles/mm³ concentration. (b) A sample of instantaneous velocity vector field measured using our DIH-PTV and (c) the corresponding structured vector field superimposed with the contours of the streamwise velocity magnitude. (d) A comparison of a sample of instantaneous streamwise velocity profile measured from DIH-PTV and Poiseuille profile in the channel. The instantaneous profile is spatially averaged over a region of 50 μ m thick and 100 μ m in length around the xz -middle plane and (e) the distribution of corresponding longitudinal velocity component (w).

3.2.2 Case #2: application to DNS turbulent channel flow

Synthetic tracer field in a turbulent channel flow is constructed using the DNS database from Johns Hopkins University [26,27]. Specifically, an initial tracer field is generated with concentration of 2200 particles/mm³ uniformly spaced within a $512 \times 512 \times 1024 \mu\text{m}^3$ volume of interest. Then the generated tracer field is advected using the 3D velocity field extracted from the DNS channel flow database to produce consecutive tracer fields in time. Based on a similar approach used in the previous section, the corresponding synthetic holograms are generated from these tracer fields, simulating red light illumination and $1 \mu\text{m}/\text{pixel}$ resolution for particles of $2 \mu\text{m}$ in diameter. Subsequently, the tracer fields are extracted through our IIPE algorithm with more than 97% extraction rate and less than $3 \mu\text{m}$ positioning error for successively simulated holograms. Following the procedure explained in Case #1, the unstructured 3D velocity fields are computed for every two successively recorded holograms. The 3D displacement fields from the exact (DNS) and measured (DIH-PTV) are presented in Fig. 8. Evidently, the DIH-PTV results demonstrate a good replication of the exact 3D velocity field from DNS. The displacement uncertainty defined similar to the positioning error δ , i.e. maximum error for 95% of extracted vectors, is $< 8\%$ in magnitude.

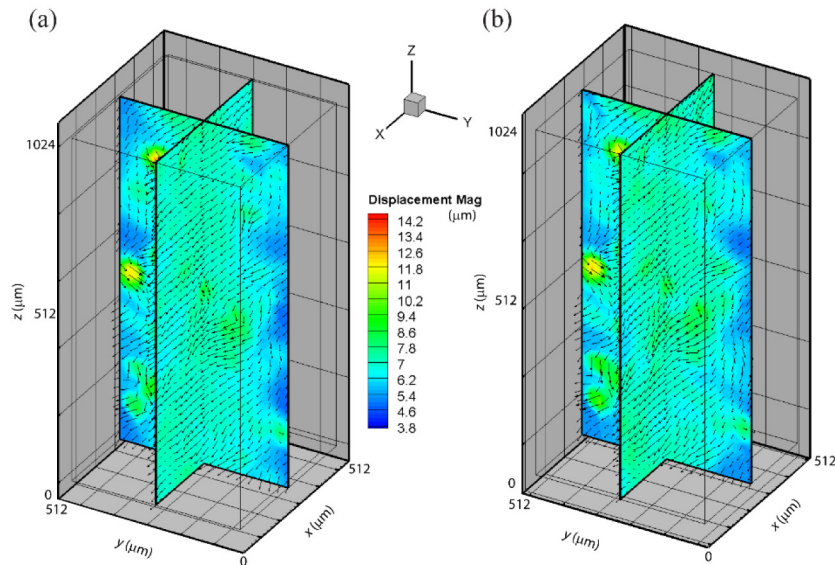


Fig. 8. (a) A sample of instantaneous 3D displacement vector field from JHUDNS database superimposed with its displacement magnitude contour, (b) corresponding 3D displacement vector field calculated through our DIH-PTV similarly superimposed with its displacement magnitude contour.

4. Summary and discussion

A novel DIH-PTV particle extraction method including 3D deconvolution, automatic 3D SNR enhancement and thresholding, and iterative inverse particle extraction is developed and implemented using GPU-based computational algorithm. The method is evaluated using simulated holograms consisting of concentrated synthetic particle fields with different particle concentration, particle size and volume in-depth lengths. Our method yields high accuracy of particle positioning ($\delta < 1.6 D_p$) and high extraction rate ($E_p > 93\%$) for concentrated particle fields (up to $C_p \sim 5000$ particles/mm³), making it an ideal method for high resolution 3D flow measurements. The method has also been implemented to quantify 3D flow fields in a laminar flow microchannel seeded with concentrated tracers ($C_p \sim 3000$ particles/mm³). The measured instantaneous velocity field matches the analytical Poiseuille profile with less than

5% difference. This method has been further validated using synthetic holograms replicating the DNS turbulent channel flows, and the results show less than $0.2 D_p$ in magnitude difference between the measured 3D turbulent flows from our method and the corresponding exact solutions.

Overall, our method has significantly improved the existing DIH-PTV approach by overcoming its major limitations including poor longitudinal resolution, human intervention and limited tracer concentration and expensive computations. Such improvements could substantially enhance the implementation of DIH-PTV for 3D flow measurements and enable the potential commercialization of this technique. Moreover, our achievement has also broadened the range of DIH applications to metrology and other *in situ* measurements in highly concentrated suspensions and natural flows for characterizing 3D motion and concentration. It is also worth noting that future improvements on deconvolution algorithm, inverse iterative particle extraction scheme as well as GPU-based computation can potentially push the boundary of our method for sub-pixel resolution particle extraction with concentration above 10^5 particles/mm³ in real-time.

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